

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil)
```

```
import pandas as pd
```

```
import numpy as np
```

```
print(pd.__version__)
```

```
2.2.2
```

```
#from list
ser=pd.Series([1,2,3,4,5],index=['a','b','c','d','e'])
```

```
print(ser)
```

```
a    1
b    2
c    3
d    4
e    5
dtype: int64
```

```
student={
    "name":"vamshi",
    "branch":"aiml"
}
se=pd.Series(student)
print(se)
```

```
name      vamshi
branch    aiml
dtype: object
```

```
n=pd.Series([10,20,30])
```

```
n*2
```

```

  0
0  20
1  40
2  60

dtype: int64
```

```
n-1
```

```

      0
0    9
1   19
2   29

dtype: int64

```

```

n.mean()
n.sum()
n.max()
n.describe()

```

```

      0
count    3.0
mean    20.0
std     10.0
min     10.0
25%     15.0
50%     20.0
75%     25.0
max     30.0

dtype: float64

```

```
n[n>20]
```

```

-----
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_1634/874502643.py in <cell line: 0>()
----> 1 n[n>20 and n<40]

/usr/local/lib/python3.12/dist-packages/pandas/core/generic.py in __nonzero__(self)
    1575     @final
    1576     def __nonzero__(self) -> NoReturn:
-> 1577         raise ValueError(
    1578             f"The truth value of a {type(self).__name__} is ambiguous. "
    1579             "Use a.empty, a.bool(), a.item(), a.any() or a.all()."

ValueError: The truth value of a Series is ambiguous. Use a.empty, a.bool(), a.item(), a.any() or
a.all().

```

Next steps: [Explain error](#)

```

names=pd.Series(['vamshi','pushpa','ram'])
names.str.lower()

```

```

      0
0  vamshi
1  pushpa
2    ram

dtype: object

```

```
names.str.len()
```

```

0
0 6
1 6
2 3

```

dtype: int64

```
names.str.startswith('A')
```

```

0
0 False
1 False
2 False

```

dtype: bool

```
names.str.upper()
```

```

0
0 VAMSHI
1 PUSHPA
2 RAM

```

dtype: object

```

students={
    "names":["vamsi","ram","jayanthi","ashwitha","bunny"],
    "branch":["aiml","aiml","aiml","aiml","aiml"]
}
s=pd.DataFrame(students)
print(s)

```

```

      names branch
0    vamsi   aiml
1      ram   aiml
2 jayanthi   aiml
3 ashwitha   aiml
4    bunny   aiml

```

```

std=[
    {"name":"vamsi","branch":"aiml"},
    {"name":"ram","branch":"aiml"},
    {"name":"jayanthi","branch":"aiml"},
    {"name":"ashwitha","branch":"aiml"},
    {"name":"bunny","branch":"aiml"}
]
s=pd.DataFrame(std)
print(s)

```

```

      name branch
0    vamsi   aiml
1      ram   aiml
2 jayanthi   aiml
3 ashwitha   aiml
4    bunny   aiml

```

```
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=['A','B','C','D'])
print(c)
```

	A	B	C	D
0	0.696495	0.677670	0.490159	0.408241
1	0.244650	0.365746	0.747669	0.794033
2	0.200879	0.045087	0.610297	0.614761

```
data={
    'Name':['Alice','Bob','Charlie'],
    'Age':[25,30,35],
    'City':['New York','San Francisco','Los Angeles']
}
d=pd.DataFrame(data)
print(d)
```

	Name	Age	City
0	Alice	25	New York
1	Bob	30	San Francisco
2	Charlie	35	Los Angeles

```
d.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  ---
0   Name    3 non-null    object
1   Age      3 non-null    int64
2   City     3 non-null    object
dtypes: int64(1), object(2)
memory usage: 204.0+ bytes
```

```
d.describe()
```

	Age	
count	3.0	
mean	30.0	
std	5.0	
min	25.0	
25%	27.5	
50%	30.0	
75%	32.5	
max	35.0	

```
d.index
```

```
RangeIndex(start=0, stop=3, step=1)
```

```
d.dtypes
```

```
0
Name  object
Age   int64
City  object

dtype: object
```

```
d.nunique()
d.isnull().sum()
```

```
0
Name  0
Age   0
City  0

dtype: int64
```

```
d["Name"]
```

```
   Name
0  Alice
1   Bob
2  Charlie

dtype: object
```

```
d.Name
```

```
   Name
0  Alice
1   Bob
2  Charlie

dtype: object
```

```
d[['Name', 'Age']]
```

```
   Name  Age
0  Alice  25
1   Bob  30
2  Charlie 35
```

```
d.iloc[0]
```

```

      0
Name    Alice
Age      25
City    New York

```

dtype: object

```
d.loc[0:2, "Name": "Age"]
```

```

      Name  Age
0    Alice  25
1     Bob  30
2  Charlie  35

```

```
d.iloc[0]
```

```

      0
Name    Alice
Age      25
City    New York

```

dtype: object

```
d.iloc[0:2]
```

```

      Name  Age      City
0    Alice  25    New York
1     Bob  30  San Francisco

```

```
d.iloc[0:2, 0:2]
```

```

      Name  Age
0    Alice  25
1     Bob  30

```

```
d.iloc[-1]
```

```

      2
Name    Charlie
Age      35
City    Los Angeles

```

dtype: object

```
d.iloc[0,0]
```

```
'Alice'
```

```
d.iloc[0, 1]
```

```
u.info['name']
```

```
'Alice'
```